

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 7: Class A Take-Off Flight Path & 4 Climb Segments

Syllabus Source: CAE Oxford (pp. 239-282) | Pass Target: >= 90% AviationExam

1. THE NET FLIGHT PATH & THE 4 CLIMB SEGMENTS

The take-off flight path begins at 35 ft screen height (dry) or 15 ft (wet) with one engine inoperative, continuing to 1,500 ft AGL. It is divided into four distinct aerodynamic segments:

| Climb Segment              | Physical Boundaries & Speeds                                                        | Aircraft Configuration & Min Gross Gradient (Tw                                                                  |
|----------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Climb Segment              | Physical Boundaries & Speeds                                                        | Aircraft Configuration & Min Gross Gradient (Twins)                                                              |
| 1st Segment                | 35 ft screen height to Gear Retraction fully complete. Speed = V2.                  | Take-off flaps, gear extending/retracting, Take-Off Thrust (TOGA). Min Gross Gradient: Positive (>= 0.0%).       |
| 2nd Segment (CRITICAL!)    | Gear Retracted up to Acceleration Height (min 400 ft AGL). Speed = V2.              | Take-off flaps, gear UP, TOGA thrust. Min Gross Gradient: 2.4% for twins (3.0% for quads) -- THE MOST STRINGENT! |
| 3rd Segment (Acceleration) | Level acceleration flight at min 400 ft AGL. Speed accelerates from V2 to V_FTO.    | Flaps retracting sequentially, gear UP, TOGA thrust. Min Gross Gradient: Acceleration equivalent.                |
| 4th Segment (Final)        | From Acceleration Height to min 1,500 ft AGL. Speed = V_FTO (Final Take-Off Speed). | Clean configuration (flaps UP), gear UP, Max Continuous Thrust (MCT). Min Gross Gradient: 1.2% for twins.        |

2. GROSS VS NET TAKE-OFF FLIGHT PATH

Obstacle clearance calculations are NOT based on actual test flight performance (Gross):

- Gross Flight Path: The actual climb performance achieved by a skilled test pilot in a brand-new aircraft.
- Net Flight Path (EASA Deduction): Gross flight path reduced by a mandatory regulatory penalty to account for engine wear and pilot technique: Deduct 0.8% for twin-engine jets! (Deduct 0.9% for 3-engines, 1.0% for 4-engines).
- Obstacle Clearance Margins: The Net Flight Path must clear all ground obstacles by at least: 35 ft vertically in straight flight, or 50 ft vertically during turns (max bank angle 15°)! The wings must clear terrain horizontally by at least 90 m (or 300 m if radar tracking unavailable).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

**[\*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Net Flight Path Obstacle Clearance**

SCENARIO (Classic CS-25 Obstacle Clearance Problem): A twin-engine jet takes off with an engine failure at V1: - Screen height (35 ft = 10.7 m) reached at end of TODA - In the 2nd segment, the aircraft achieves a Gross Climb Gradient of 2.8% - An obstacle of height 180 ft (55.0 m) is located 3,000 meters beyond the end of TODA QUESTION: What is the NET climb gradient, and what is the aircraft's NET clearance over the obstacle? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the NET Climb Gradient (Deduct 0.8% for twins!): - Gross Gradient = 2.8% - Net Penalty (CS-25 twin) = -0.8% - Net Gradient = 2.8% - 0.8% = 2.0% (0.020). Step 2: Calculate the Net Altitude Gained over the 3,000 m distance to obstacle: - Net Altitude Gained = Horizontal Distance x Net Gradient - Net Altitude Gained = 3,000 m x 0.020 = 60.0 meters. Step 3: Calculate the Net Aircraft Height at the obstacle position: - Net Height = Initial Screen Height (10.7 m) + Net Altitude Gained (60.0 m) - Net Height = 10.7 m + 60.0 m = 70.7 meters (approx 232 ft). Step 4: Check Clearance Margin above the Obstacle: - Obstacle Height = 55.0 meters (180 ft) - Net Clearance = Net Aircraft Height - Obstacle Height - Net Clearance = 70.7 m - 55.0 m = 15.7 meters (51.5 ft)! Step 5: Verify Legal Margin (Must be  $\geq$  35 ft): - Since 51.5 ft > 35 ft, the aircraft legally clears the obstacle! FINAL ANSWER: Net Climb Gradient = 2.0%. Net Clearance = 15.7 m (51.5 ft), which comfortably exceeds the 35 ft legal requirement.

**[!] EXAM TRAP (AVIATIONEXAM) - The 2nd Segment Gradient Trap**

- The 2nd segment is the MOST CRITICAL climb segment in ATPL exams! - Twin-engine minimum gross gradient = 2.4% (Net = 1.6%). - If high elevation or hot temperatures degrade climb below 2.4%, Take-Off Mass MUST be reduced, even on an infinitely long runway!

**4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**

**Q1: Under CS-25, what is the mandatory Gross to Net gradient deduction for a TWIN-ENGINE transport aircraft?**

[A] 0.5% [B] 0.8% gradient reduction [C] 1.0% [D] 1.5%

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. CS-25 mandates a gradient penalty of 0.8% for 2-engine aircraft, 0.9% for 3-engine aircraft, and 1.0% for 4-engine aircraft to convert gross flight paths into legal net flight paths.

**Q2: What is the minimum required GROSS climb gradient in the SECOND SEGMENT for a twin-turboprop aircraft?**

[A] 0.0% [B] 2.4% gross climb gradient [C] 1.2% [D] 3.0%

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. In the second segment (gear up, flaps take-off, V2, TOGA thrust), a twin must achieve at least a 2.4% gross climb gradient (yielding a 1.6% net gradient).

**Q3: What is the minimum vertical obstacle clearance required between the NET take-off flight path and an obstacle?**

[A] 50 ft [B] 35 ft vertical clearance [C] 100 ft [D] Zero

**[\*] OFFICIAL DEFINITION - Detailed Answer Explanation**

CORRECT: [B]. CS-25 requires the net take-off flight path to clear all obstacles by at least 35 ft vertically in wings-level flight (increased to 50 ft during turns).